

SCIENCE IS THE PATH TO KNOWLEDGE, AND FINDING OUT ABOUT HOW OUR UNIVERSE WORKS. YOU CAN'T BE A GEEK AND NOT KNOW YOUR SCIENCE!

THIS MONTH IN SCIENCE:

An aurora-like phenomenon called Steve, the origins of surgery, 2D materials and what they're good for, the mysteries of alkaloids, very large telescopes, what big data does for health, and electric cars that see better...



Bottlenecks in the brain

Researchers from Carnegie Mellon University have discovered that there are bottlenecks between portions of the brain, that regulate the information flow between the neurons these areas. These communication subspaces decide where some information, such as colour or motion, is routed to. The discovery could better help explain brain activity.
<https://digit.in/nbtink>

Ancient dino cousin shows diversification in post extinction Antarctica

An iguana-sized cousin of the dinosaurs and modern crocodilians has been discovered, in the *Antarctanax* genus, which means "Antarctic King". The creature has been named *Antarctanax shackletoni*, in honour of Ernest Shackleton, a polar explorer. About 250 million years ago, Antarctica was not a frozen desert, and had a vibrant ecosystem with forests and rivers. The creature was a carnivore that preyed on insects, amphibians and relatives of the early mammals. The discovery is the latest of many new life forms found in ancient Antarctica, which are surprisingly very different from the



fossils found in Africa. The two continents were joined during the time period, in a supercontinent called

Pangea. The discovery also points to the recovery of life after the Permian-Triassic extinction, where over 90 percent of the life on Earth was wiped out because of climate change caused by massive volcanic eruptions in the Siberian traps.

<https://digit.in/arcking>

Neptune's newest moon may be a piece of another moon

In 2013, Hubble photographed a tiny moon of Neptune, which was given the name Hippocamp, after a creature from Greek mythology. The object puzzled astronomers as the moon simply should not have been there. It is only about 12,000 kilometers away from Proteus, another moon of Neptune. At such a distance, Proteus



should have absorbed the tiny moon, in a process by which large objects clear tinier objects in their orbital path.

Astronomers looking through old photos, captured by the Voyager 2 spacecraft in 1989, noticed a large crater on Proteus. The crater was evidence that a cometary impact could have chipped off Hippocamp from Proteus. This is the first time that there is evidence that a moon was formed after a cometary impact event.

<https://digit.in/hpcmp>



A piper piper for brain tumors

In a breakthrough for treating brain tumours, researchers have created a "Tumor Monorail" that makes the tumours migrate to the outer regions of the brain, where they can be removed easily. The prototype device mimics white matter within brains.
<https://digit.in/ppbrain>



Rocket launched from balloon

A startup by the name of Leo Aerospace has successfully launched a rocket into space, from a high altitude balloon. The approach could make space more accessible to private companies, and allow for the deployment of microsatellites at low costs.
<https://digit.in/leospce>



Vegetarian diet may control diabetes

Studies show that those who have a vegetarian diet have healthier blood sugar levels, a lower body weight, and reduced dependence on medication. Even just reducing the amount of meat intake, can have beneficial effects.
<https://digit.in/vegdb>